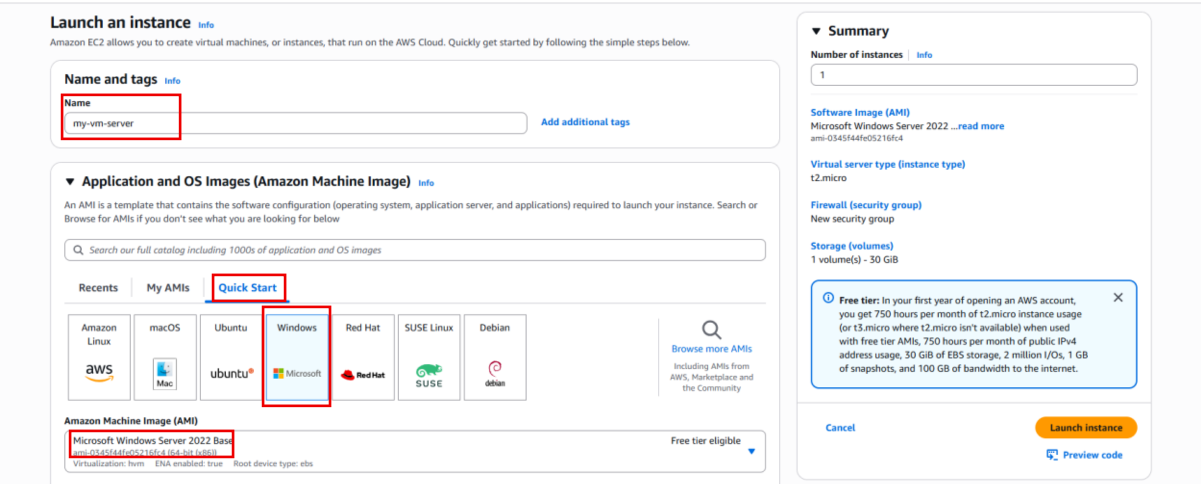
Replace the root volume for an Amazon EC2 instance without stopping it

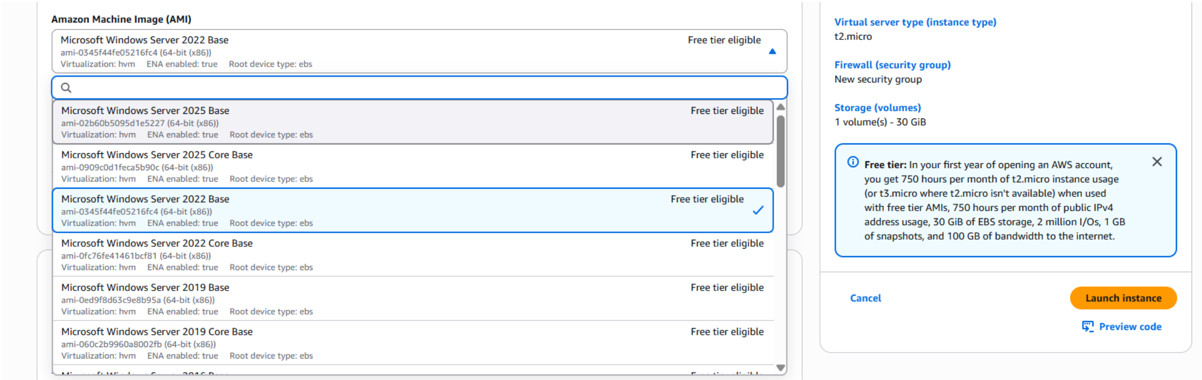
To begin with the lab

Summary of the Lab

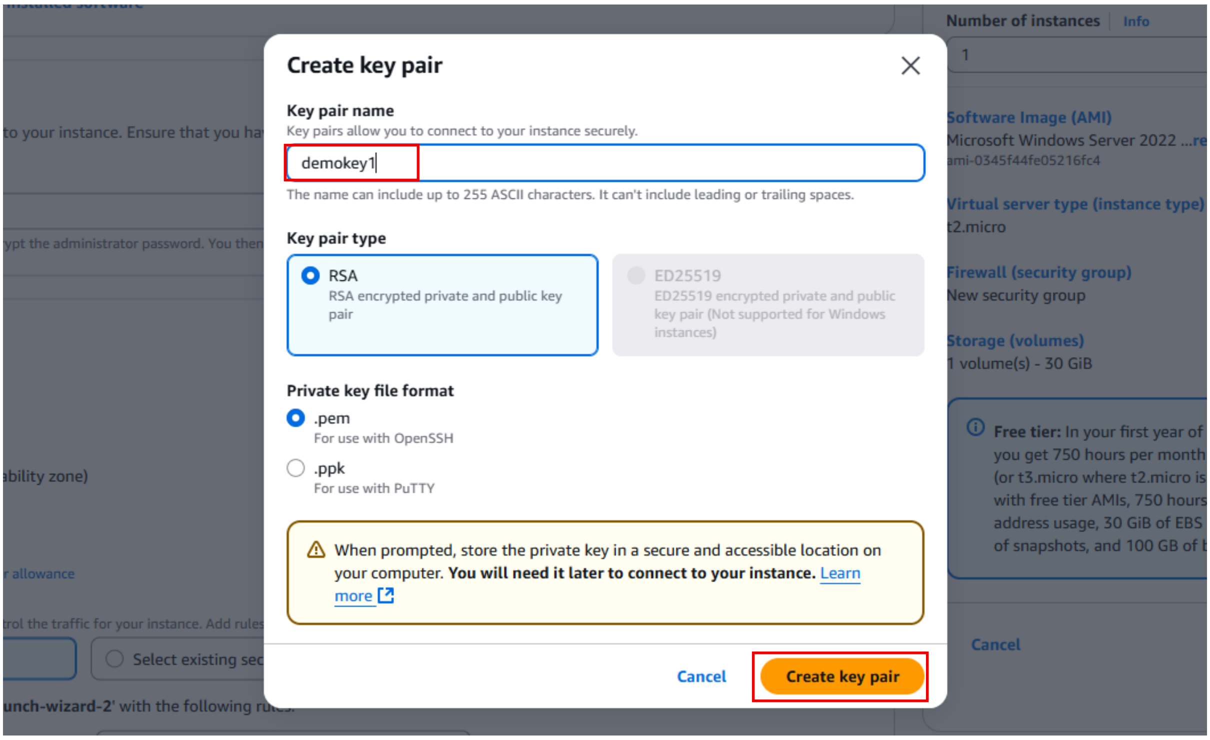
To create and work with a Windows Server 2022 EC2 instance, launch a new instance on AWS, set up a key pair and security group, and connect via RDP using the decrypted password. Inside the VM, create a new folder and a text file in the C: drive. Then, go to AWS Console and create a snapshot from the volume. After deleting the folder from the VM, replace the root volume using the snapshot. AWS will stop the instance, replace the root volume, and restart it. Reconnect to the VM to verify that the folder has been successfully restored.

- Let's create an instance first.
- Go to AWS Console.
- Navigate to EC2.
- Click on launch an instance.
- Name you instance.
- In Application and OS Images (Amazon Machine Image), go to quick start
- Select Windows (Microsoft)
- In Amazon Machine Image (AMI), Select Microsoft Windows Server 2022 Base.

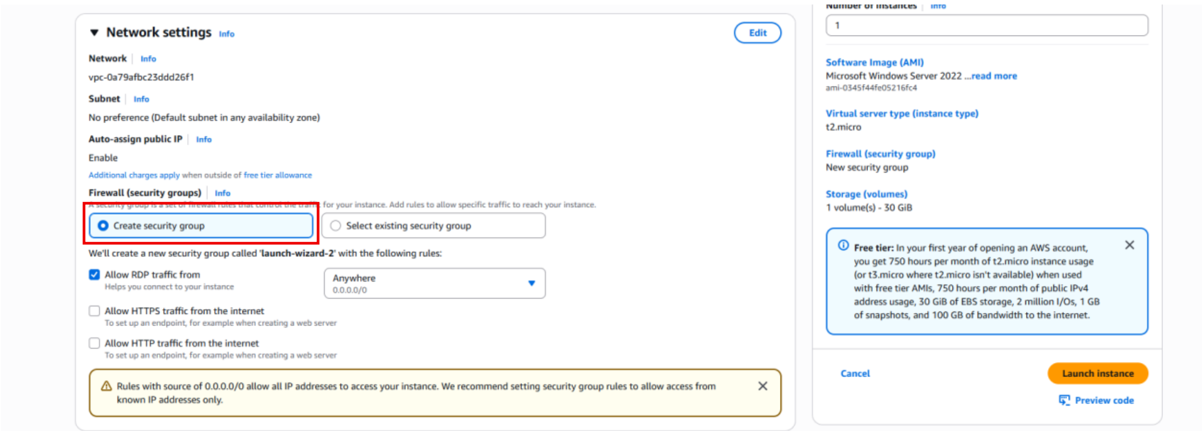




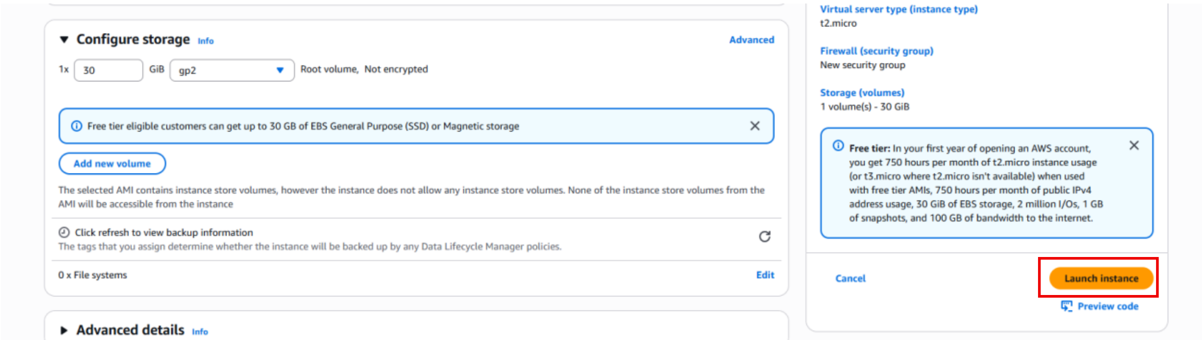
- Now in key pair (login)
- If you have existing key pair, you can use that.
- Otherwise, click on the ‘Create new key pair’.
- Name your key pair.
- Click on the ‘create key pair’
- Choose the options as given.



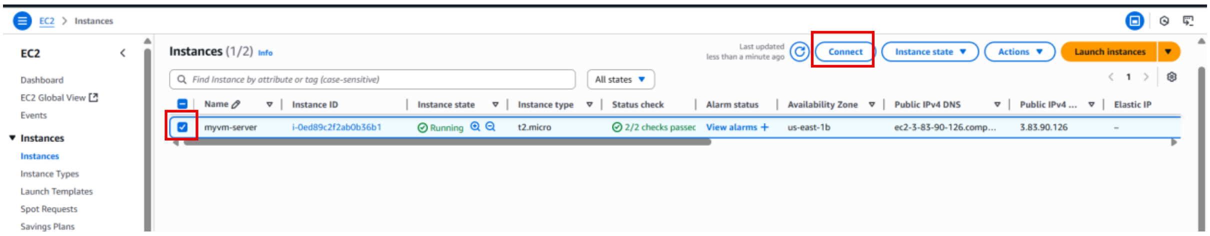
- It will download a ‘.pem’ file that will used to login to the virtual machine.
- Go to Network settings
- If you have existing security group, select that.
- Otherwise, click on the create security group
- Choose the option given below.



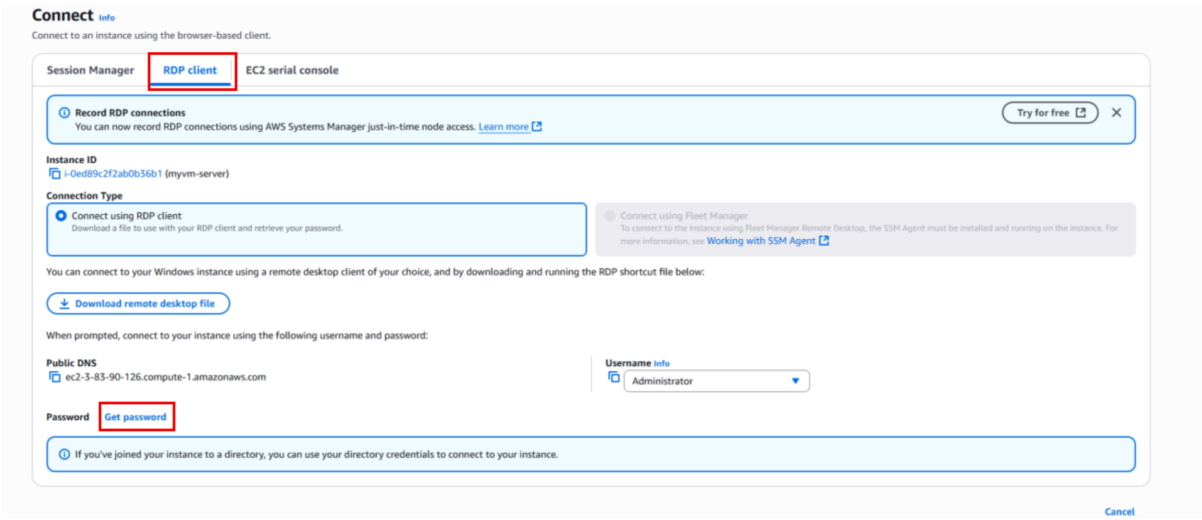
- For now, I am not increasing the memory size for the root volume.
- You can increase according to your need.
- Click on the launch instance.



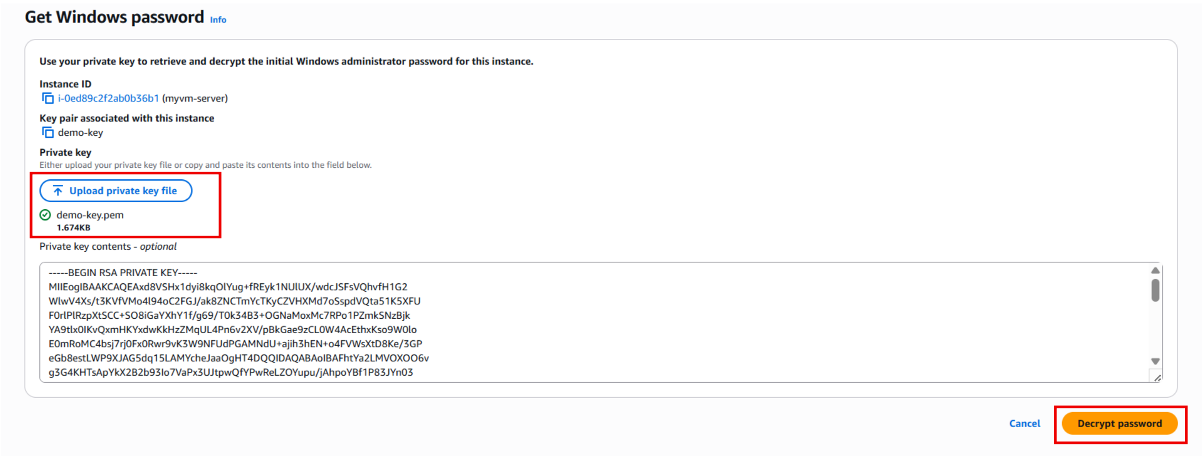
- Here is the new created instance.
- Select the instance.
- Click on the connect.



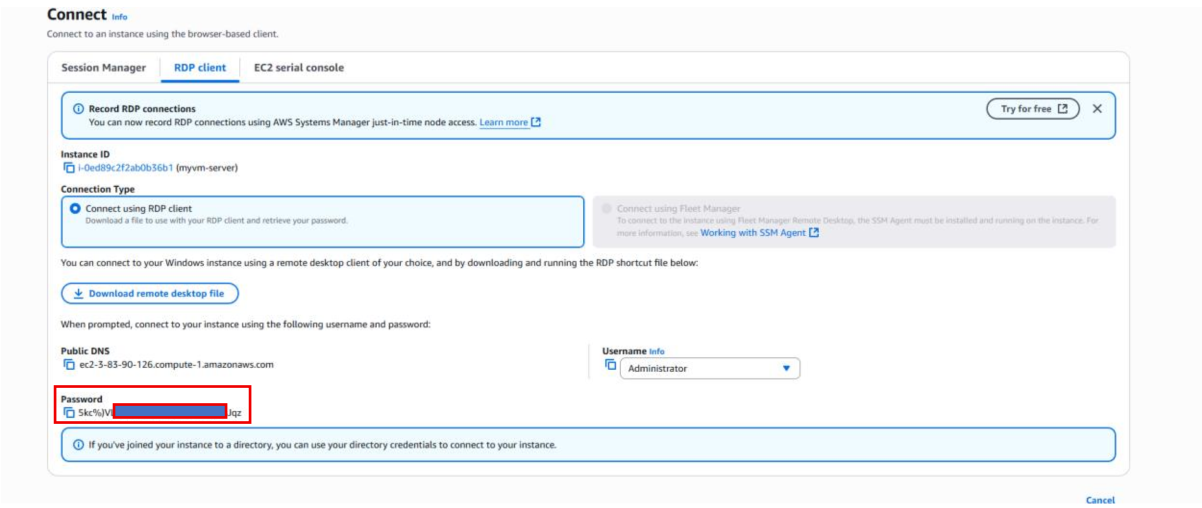
- Now in connect, go to RDP client.
- Either download the remote desktop file.
- Click on the 'get password'.



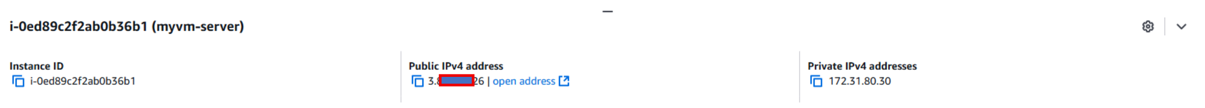
- In the 'get window password'
- Upload the private key file. (upload the key(.pem) that was downloaded.
- Click on the 'Decrypt password'



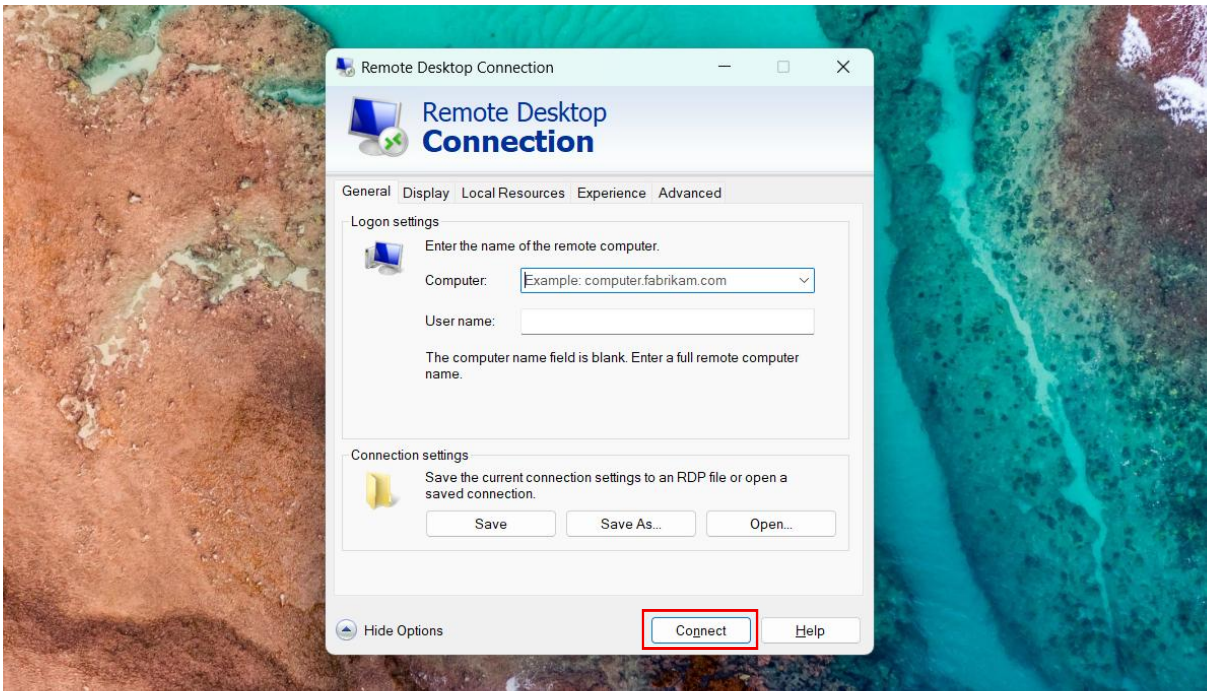
You will get you password.

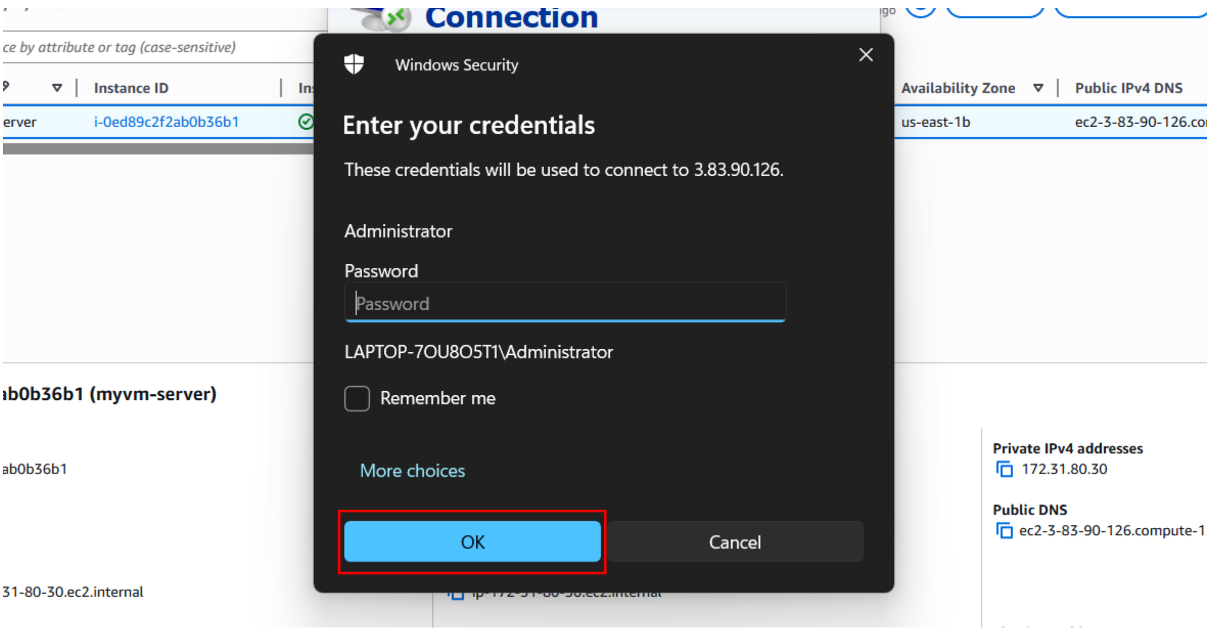


- Now copy the Username, password and the public IPv4 address.

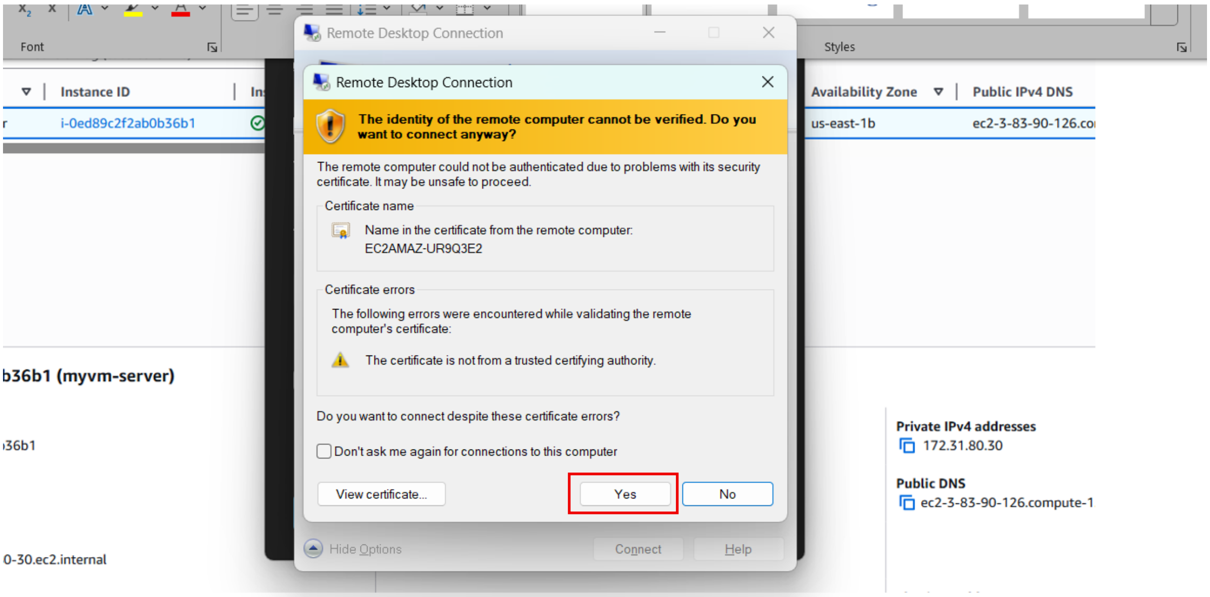


- Now search for the Remote Desktop Connection.
- In computer option paste your public IP address.
- Paste your username.
- Click on the connect.
- Then paste your password and click OK.

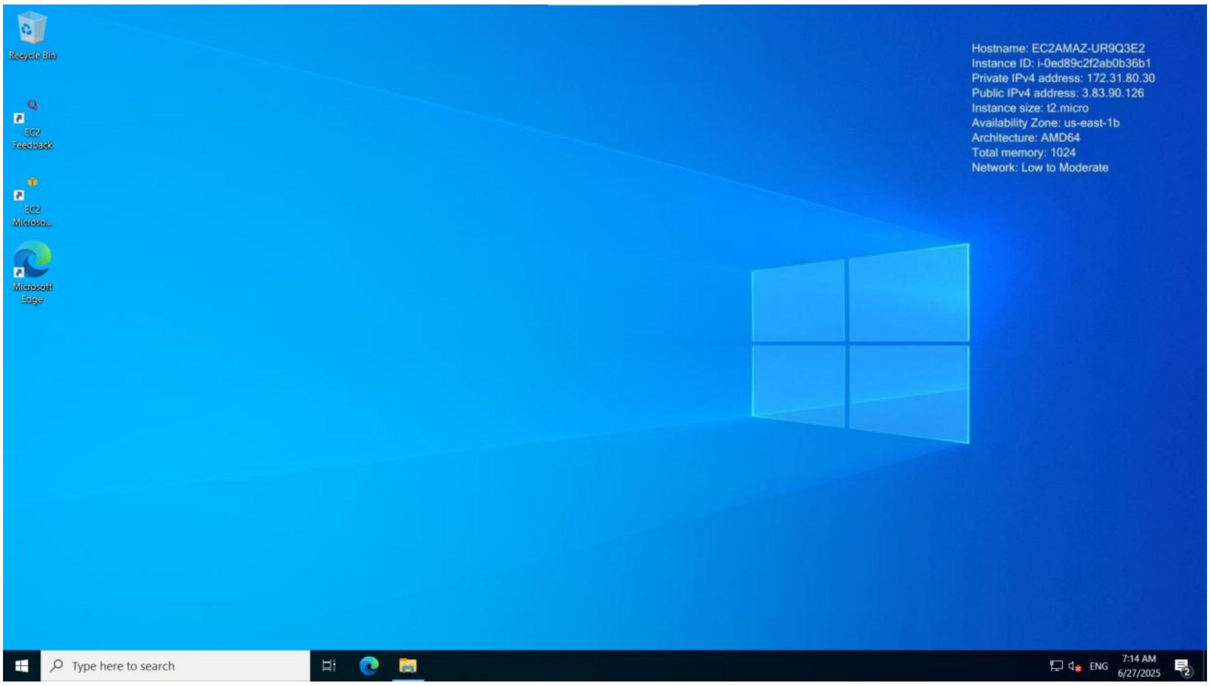




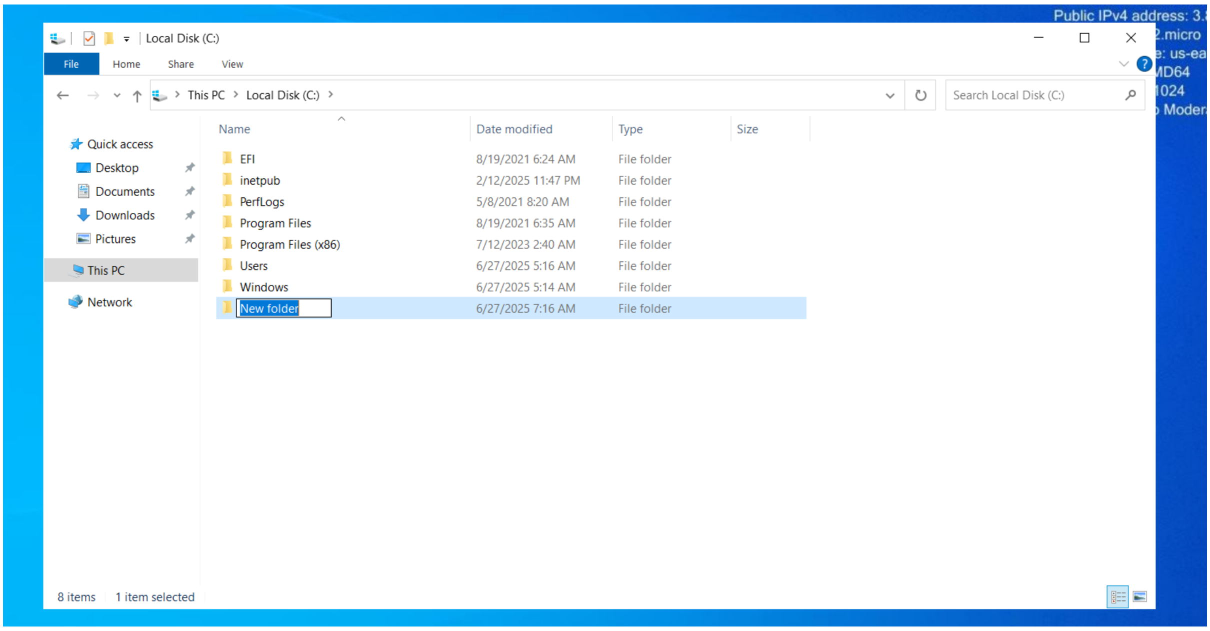
- Click OK.

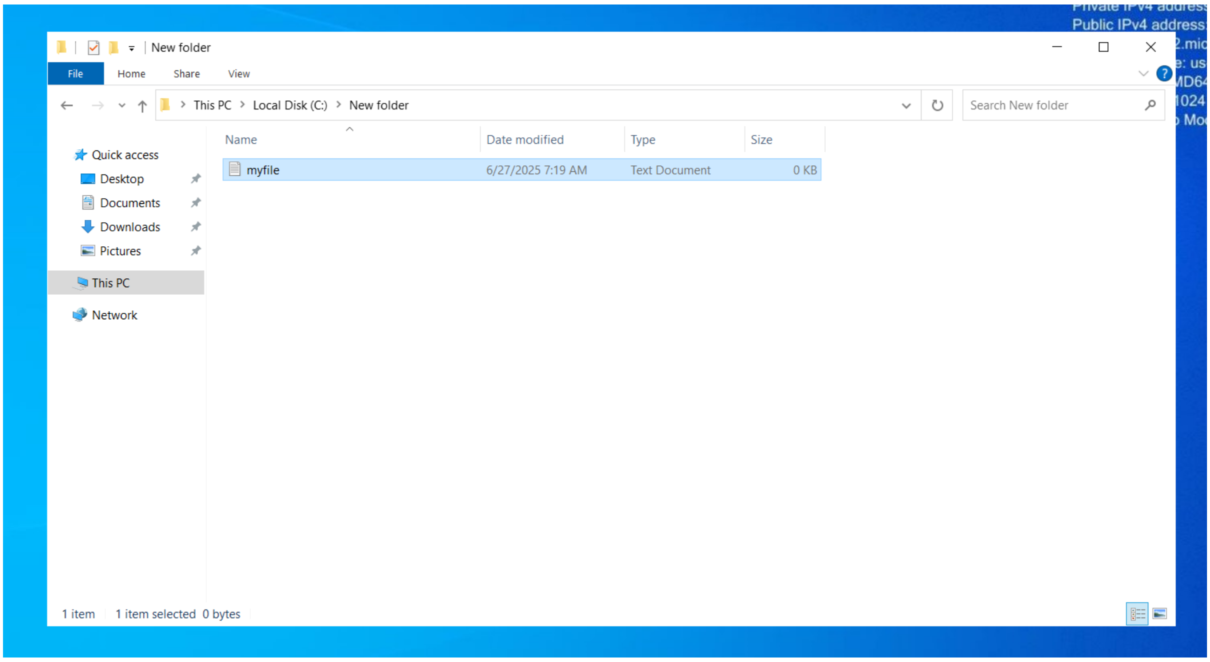


- Now we are successfully logged in to the virtual machine.

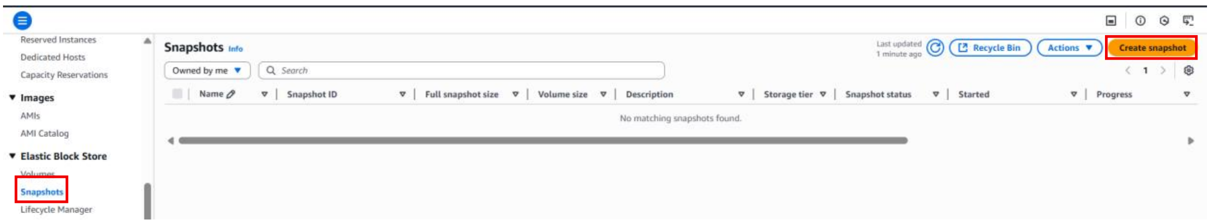


- Now here we will create a new folder in the root drive (C drive).
- And if you want to rename it you can do it.
- Now we will create a .txt file inside that folder.
- Put some content inside the file if you want to.



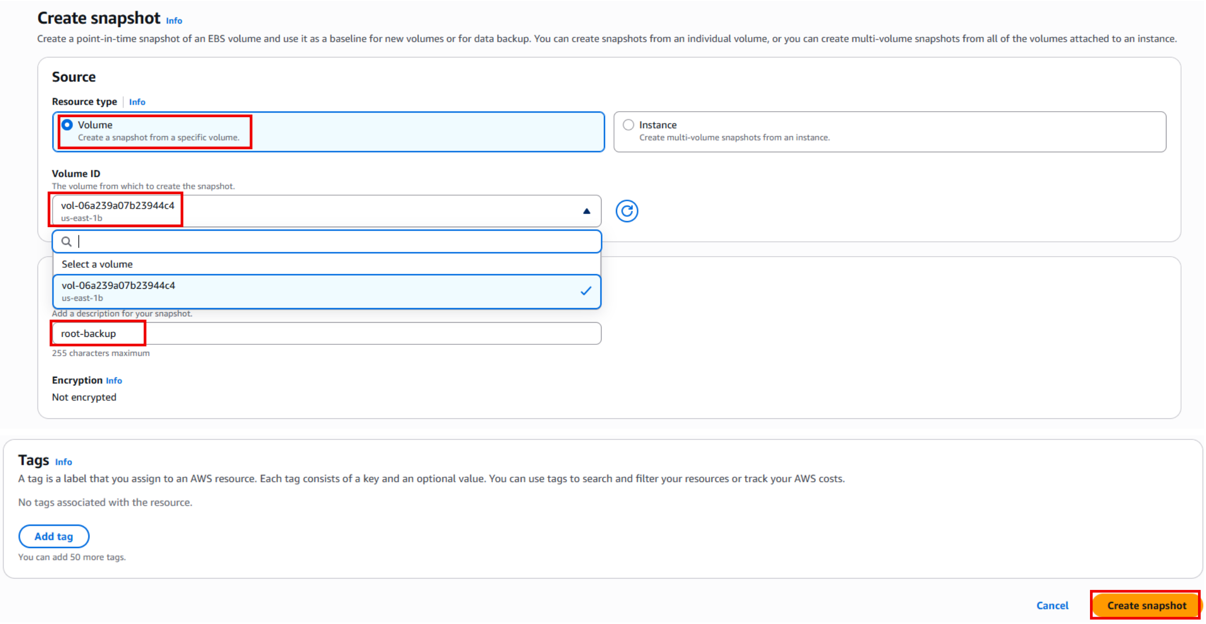


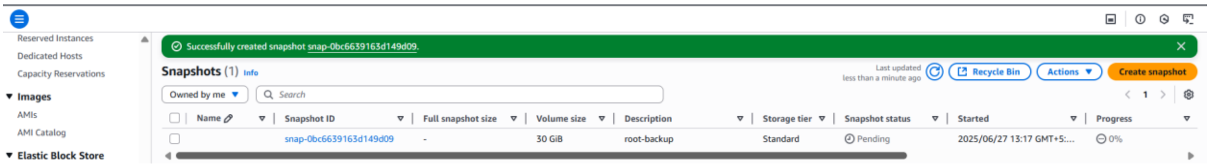
- Now let's go back to aws console.
- Go to EC2 > Snapshot.
- Create a snapshot.



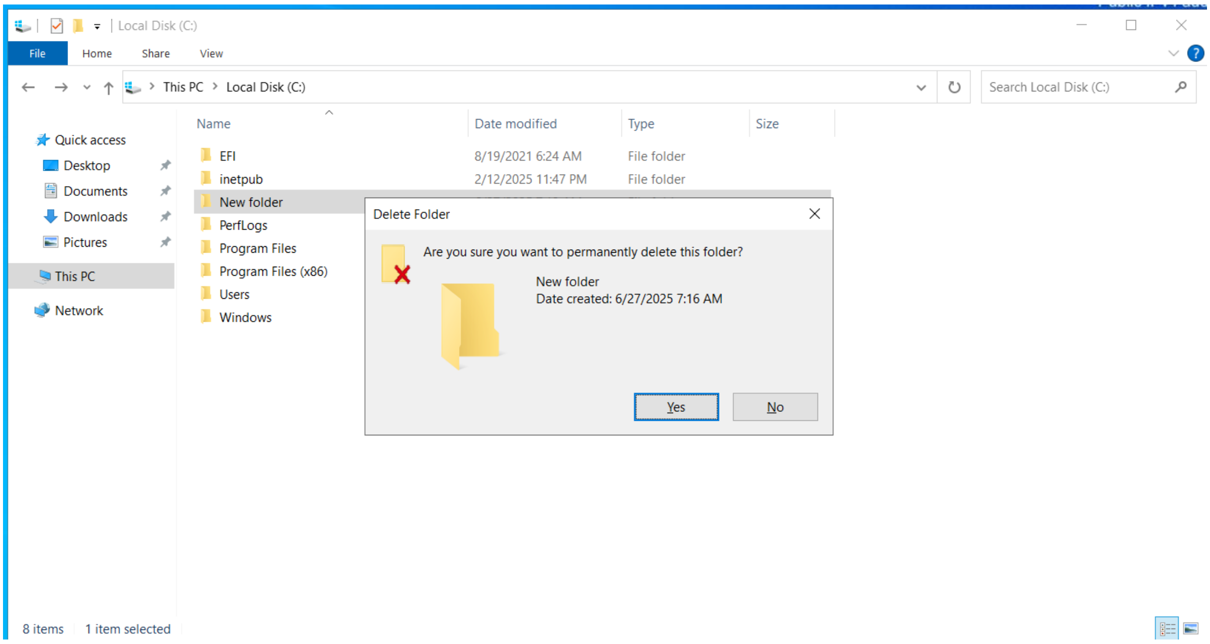
- Here we can create snapshot from the 'volume and Instance' both.
- But here I am making using volume.
- Choose the volume in the volume ID filed.

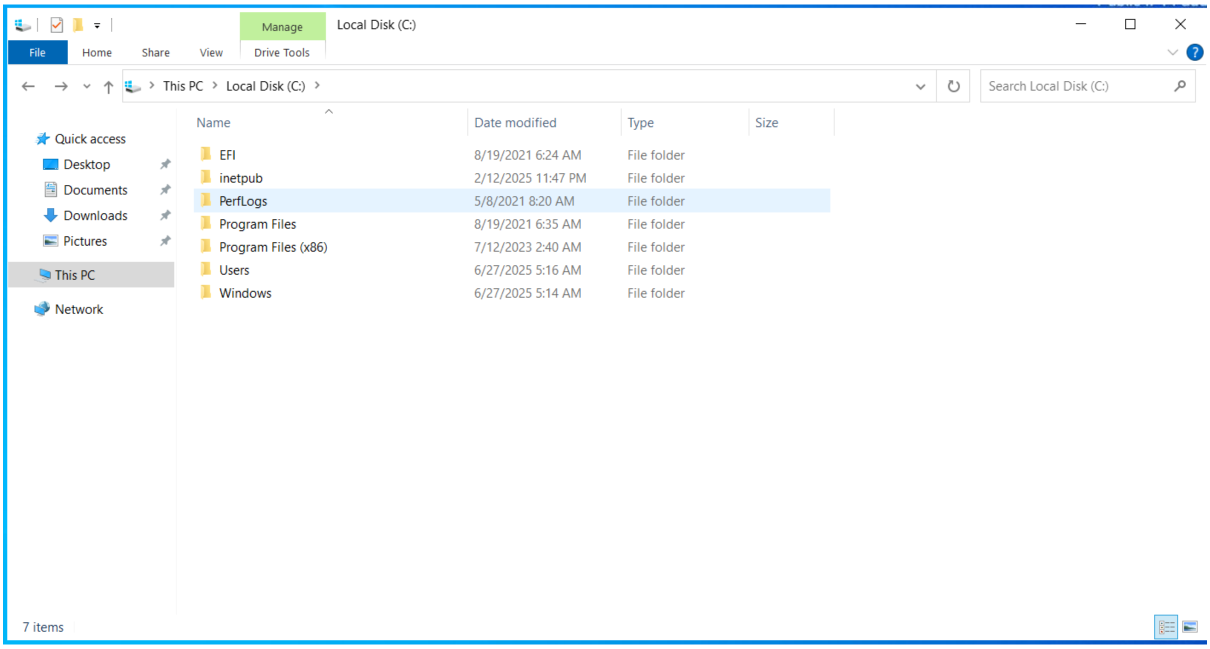
- Give the description.
- Click Create snapshot.



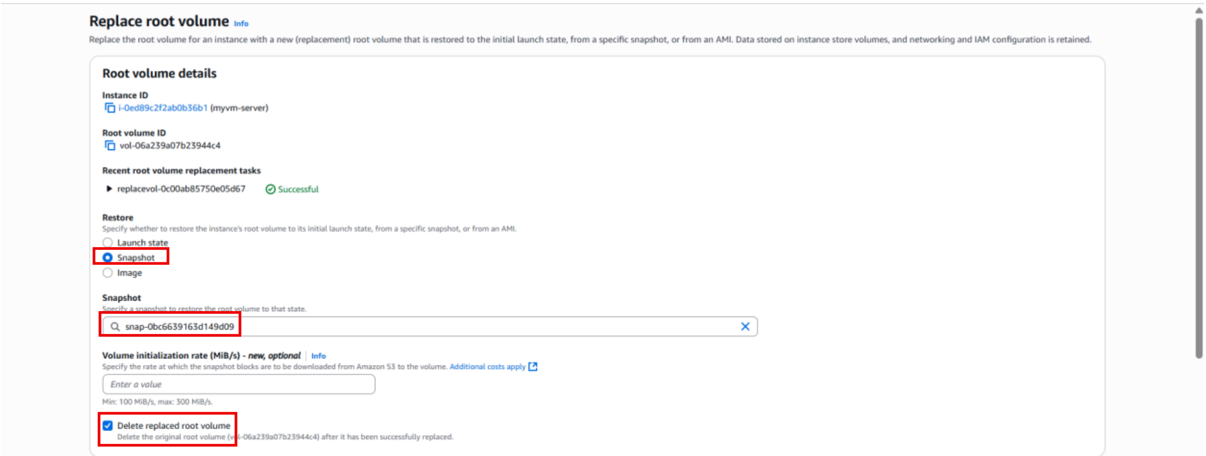


- Now I will delete the folder permanently that we have created.

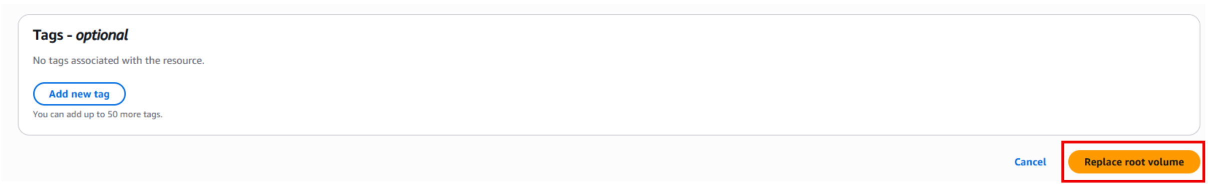




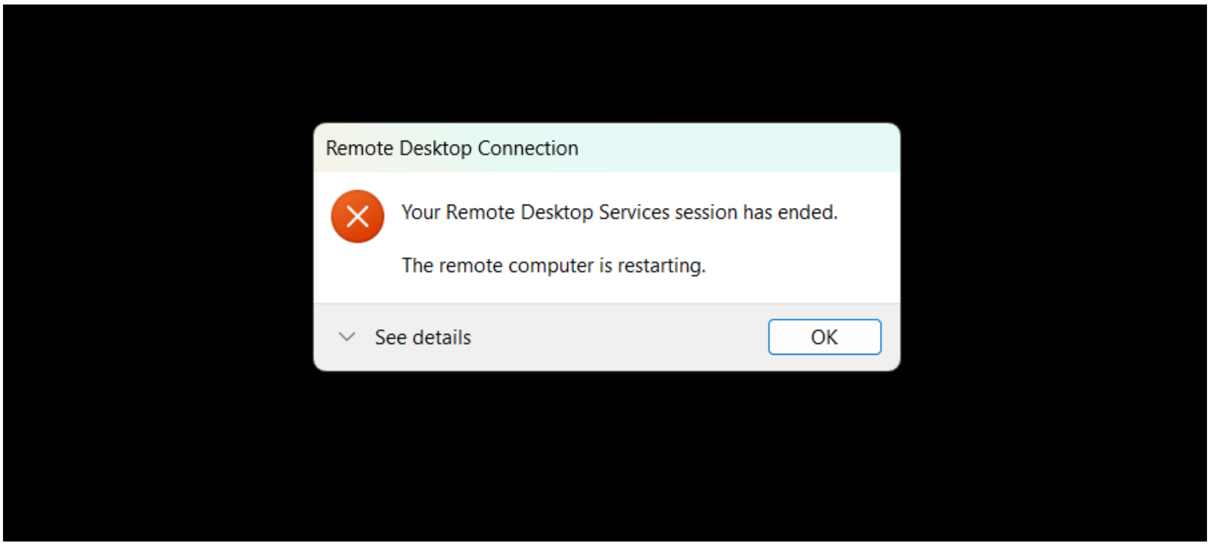
- Now go back to AWS console.
- Go to EC2 > Instance.
- Select the instance.
- Go to Action > Monitor and troubleshoot > Replace root volume.
- Choose the snapshot as an option.
- Choose the snapshot id that you have created.
- Tick the ‘delete replaced root volume’.



- Click ‘Replace root volume’.



- Your server will stop after you click replace root volume.



- Now log in again in the virtual machine.
- You can see we have successfully removed our folder.